DSP · MOCK

DSP mock B

Practice questions · Day 24 · Mon 16 Nov · pack DX2

Instructions

- Time allowed: **2 hours**. Sit it in one go, timed, with your printed notes, Test 1 Reference and reference sheet beside you (open book, hard copy only).
- Attempt **all 11 questions**. There are **60 marks** in total. Marks for each part are given in brackets. Budget about 2 minutes per mark, and keep the last 5 minutes to check.
- Show your working. Most marks go to method, so a wrong number with a correct method still scores.
- Some parts say "assuming the answer to part (x) is …". Use the value given there, even if your own answer is different. This stops one slip from costing you the rest of the question.
- Notation is the course's. ω is DT frequency (rad/sample), Ω is CT frequency (rad/s). $\operatorname{sinc} x = \sin x / x$. The arrow $\uparrow$ marks the $n = 0$ sample. Use $T = 1$ for every bilinear-transform calculation.
- This paper is meant to feel different from mock A. It uses the less common angles of L03–L10. Do not open the marking guide until time is up.

Question	Topic	Lectures	Marks
1	Correlation and delay estimation	L03	5
2	LCCDE with complex modes	L04, L05	6
3	DTFT shortcut evaluations	L05	5
4	Deconvolution via the DFT	L06	6
5	Overlap-save block convolution	L06	6
6	Two-sided inverse z-transform: every ROC	L07, L08	6
7	Unilateral z-transform with initial conditions	L08	6
8	Digital resonator and allpass filter	L09	6
9	Chebyshev type I order and ϵ	L10	5
10	Completing a linear-phase FIR filter	L10	5
11	Short answers	L06, L10	4
		Total	**60**

Question 1 [5 marks]: Correlation and delay estimation

A short radar pulse $x(n) = \{\underset{\uparrow}{1}, 2, -1\}$ is transmitted. The ADC samples the received echo at $F_s = 10$ MHz:

$$y(n) = \{\underset{\uparrow}{1}, 0, -1, 2, 4, -2, 1\}, \qquad y(n) = 0 \text{ elsewhere.}$$

Use the lecture definitions $r_{xx}(l) = \sum_n x(n)x(n-l)$ and $r_{yx}(l) = \sum_n y(n)x(n-l)$.

a. Compute the autocorrelation $r_{xx}(l)$ for all l. **[1]**

b. Compute $r_{yx}(l)$ for $-2 \le l \le 6$. **[2]**

c. Estimate the delay $\hat{d}$ and the target range r. Take $c = 3 \times 10^8$ m/s. **[1]**

d. Model the echo as $y(n) \approx a\, x(n - \hat{d})$ plus interference. Use your results from (a) and (b) to estimate the gain a. State the range resolution of this radar (the change in range for a one-sample change in delay). **[1]**

Question 2 [6 marks]: LCCDE with complex modes

A causal DT LTI system is described by

$$y(n) - y(n-1) + 0.5\, y(n-2) = x(n).$$

a. Find the characteristic modes in polar form. State whether the system is stable, and give your reason. **[1]**

b. Using recursion, evaluate $h(n)$ for $0 \le n \le 6$. Assume the system is initially at rest. **[1]**

c. Find a closed-form, real-valued expression for the impulse response $h(n)$. **[2]**

d. Find the steady-state output when the input is $x(n) = 4 + 2\cos\left(\frac{\pi}{2}n\right)$ for all n. **[2]**

Question 3 [5 marks]: DTFT shortcut evaluations

Consider the signal

$$x(n) = \{1, -2, \underset{\uparrow}{3}, 0, -1, -4\}$$

with DTFT $X(\omega)$. Compute each quantity below **without** finding $X(\omega)$ explicitly.

a. $X(0)$ and $\angle X(0)$. **[1]**

b. $X(\pi)$. **[1]**

c. $\displaystyle\int_{-\pi}^{\pi} X(\omega)\, e^{j2\omega}\, d\omega$. **[1]**

d. $\int_{-\pi}^{\pi} |X(\omega)|^2 \, d\omega$. **[1]**

e. $\int_{-\pi}^{\pi} \left| \frac{dX(\omega)}{d\omega} \right|^2 d\omega$. **[1]**

Question 4 [6 marks]: Deconvolution via the DFT

An unknown input $x(n)$ of length 3 passes through a known FIR channel $h(n) = \{\underset{\uparrow}{2}, 1\}$. The complete output is recorded:

$$y(n) = \{\underset{\uparrow}{4}, 0, 1, 1\}.$$

a. Explain why a 4-point DFT is enough to recover $x(n)$ exactly from $y(n)$, with no time-aliasing. **[1]**

b. Compute the 4-point DFTs $Y(k)$ and $H(k)$. **[2]**

c. Hence find $X(k)$, and use the 4-point IDFT to recover $x(n)$. **[2]**

d. Suppose the channel were instead $h(n) = \{\underset{\uparrow}{1}, 1\}$. Explain why this method then fails with a 4-point DFT. **[1]**

Question 5 [6 marks]: Overlap-save block convolution

An FIR filter $h(n) = \{\underset{\uparrow}{1}, -1, 2\}$ filters the input

$$x(n) = \{\underset{\uparrow}{1}, 2, 0, -1, 3, 1\}.$$

You must use the **overlap-save** method with 4-point circular convolutions ($N = 4$).

a. State the number of new input samples per block, L, and how many output samples of each block are discarded. **[1]**

b. Write down every 4-sample input block you need to produce the full linear-convolution output. **[1]**

c. Compute the 4-point circular convolution of each block with $h(n)$ (you may compute only the samples you keep). Assemble $y(n) = h(n) * x(n)$. **[3]**

d. For the first block, compute the two discarded samples. Explain why they differ from the true $y(-2)$ and $y(-1)$. **[1]**

Question 6 [6 marks]: Every possible sequence for each ROC

$$X(z) = \frac{1 + 11z^{-1} - 11z^{-2}}{(1 - 0.5z^{-1})(1 - 2z^{-1})(1 + 3z^{-1})}$$

a. Expand $X(z)$ in partial fractions of the form $\dfrac{A}{1 - p\,z^{-1}}$. **[2]**

b. Assume the answer to part (a) is

$$X(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{1}{1 - 2z^{-1}} - \frac{1}{1 + 3z^{-1}}.$$

List every possible ROC. For each one, give the sequence $x(n)$ and classify it as causal, anticausal or two-sided. **[3]**

c. Using the expression in (b), which ROC gives a sequence that has a DTFT? For that sequence, evaluate $x(0)$ and $x(-1)$. **[1]**

Question 7 [6 marks]: Unilateral z-transform with initial conditions

A system is described by

$$y(n) - 0.75\,y(n-1) + 0.125\,y(n-2) = x(n),$$

with input $x(n) = 3u(n)$ and initial conditions $y(-1) = 2$, $y(-2) = 4$.

a. Take the unilateral z-transform of both sides, including the initial conditions, and obtain $Y^{+}(z)$. **[2]**

b. Find $y(n)$ for $n \geq 0$. **[3]**

c. Split your answer into its zero-input and zero-state parts. Check $y(0)$ and $y(1)$ by recursion. **[1]**

Question 8 [6 marks]: Digital resonator and allpass filter

A digital resonator is to pick out a 1 kHz tone from a signal sampled at $F_s = 8$ kHz. Use the course's resonator form, with poles at $re^{\pm j\omega_0}$, $r = 0.9$, and zeros at $z = \pm 1$.

a. Write $H(z)$ with numerical denominator coefficients. State which input frequencies (in Hz) it rejects completely. **[1]**

b. Choose b_0 so that $|H(\omega_0)| = 1$, and write the difference equation. **[2]**

A first-order allpass filter in the course form $H_{ap}(z) = z^{-N}A(z^{-1})/A(z)$ has $A(z) = 1 - 0.8z^{-1}$.

c. Write $H_{ap}(z)$. Give its pole and zero, and show that $|H_{ap}(\omega)| = 1$. **[1]**

d. Find the phase $\angle H_{ap}(\omega)$ at $\omega = \pi/2$. **[1]**

e. Find the group delay at $\omega = 0$. Explain in one line why such a filter is useful. **[1]**

Question 9 [5 marks]: Chebyshev type I order and ϵ

A digital lowpass filter will be designed by applying the bilinear transformation ($T = 1$) to an analogue Chebyshev type I filter. The specifications are:

- passband ripple ≤ 0.5 dB, passband edge 1 kHz
- stopband attenuation ≥ 40 dB, stopband edge 2.5 kHz
- sampling frequency 10 kHz

a. Find ω_p and ω_s, and the prewarped analogue edges Ω_p and Ω_s. **[1]**

b. Find δ_p, δ_s and ϵ. **[1]**

c. Find the filter order N. **[2]**

d. Assume the answer to part (c) is $N = 5$. Find the attenuation (in dB) that the analogue filter actually gives at Ω_s. **[1]**

Question 10 [5 marks]: Completing a linear-phase FIR filter

$$h(n) = \{-1, \underset{\uparrow}{2}, 4, *, -2, 1\}$$

a. Find the missing sample $h(3)$ that makes this filter linear-phase. Give your reason. **[1]**

b. Write $H(\omega)$ in the form $A(\omega)\, e^{j\theta(\omega)}$, with $A(\omega)$ real. **[2]**

c. Give the phase response and the group delay. **[1]**

d. Evaluate $H(0)$ and $H(\pi)$. Could this filter be used as a lowpass filter? Explain. **[1]**

Question 11 [4 marks]: Short answers

a. Give two reasons why the ideal lowpass filter cannot be realised in practice. **[1]**

b. A signal holds two tones at 1000 Hz and 1050 Hz. It is sampled at $F_s = 8$ kHz. A student records $L = 64$ samples and zero-pads them to $N = 1024$ before taking the DFT. State the DFT bin spacing. Can the two tones be resolved? What is the minimum record length L (and duration T_0) needed? **[2]**

c. In Question 9, a student forgets to prewarp and uses $\Omega_s = \omega_s / T = \pi/2$ rad/s directly. After the bilinear transformation, at what frequency (in Hz) does the stopband edge actually land? What does this show about the bilinear map? **[1]**

End of questions